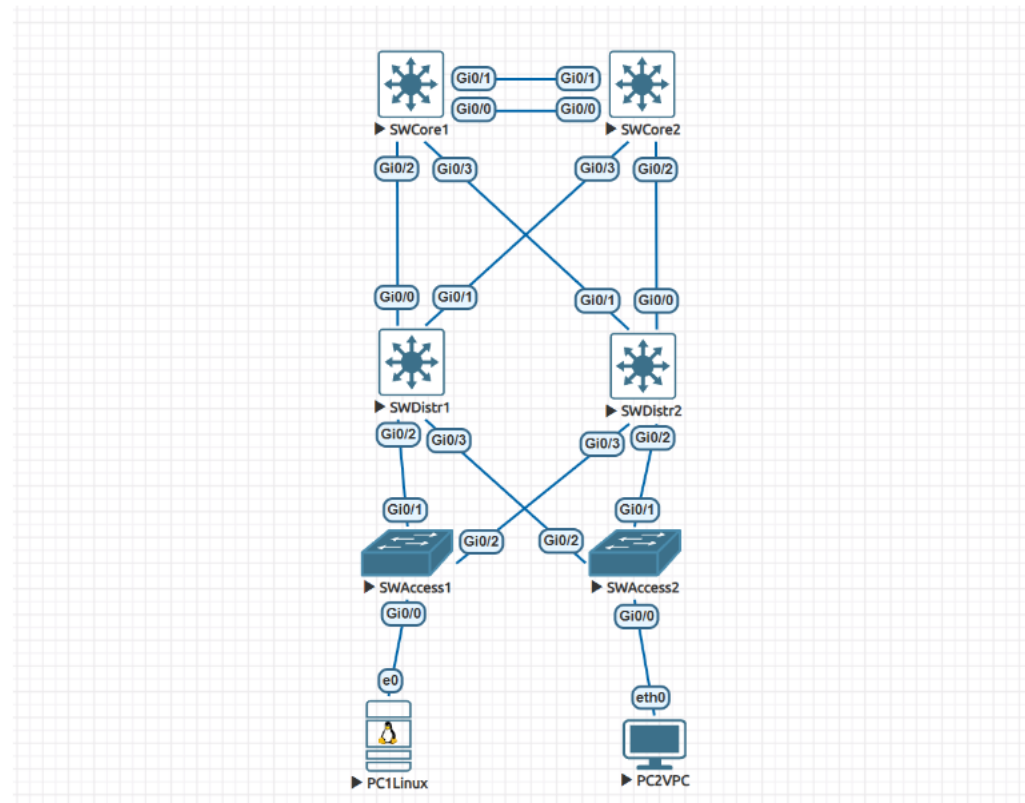
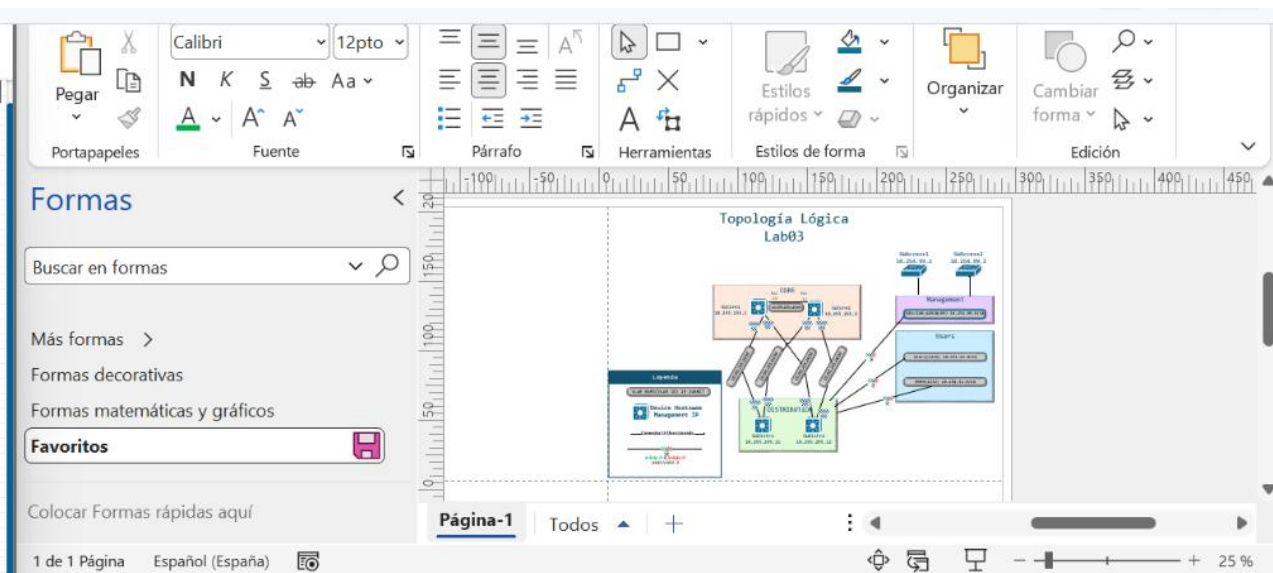
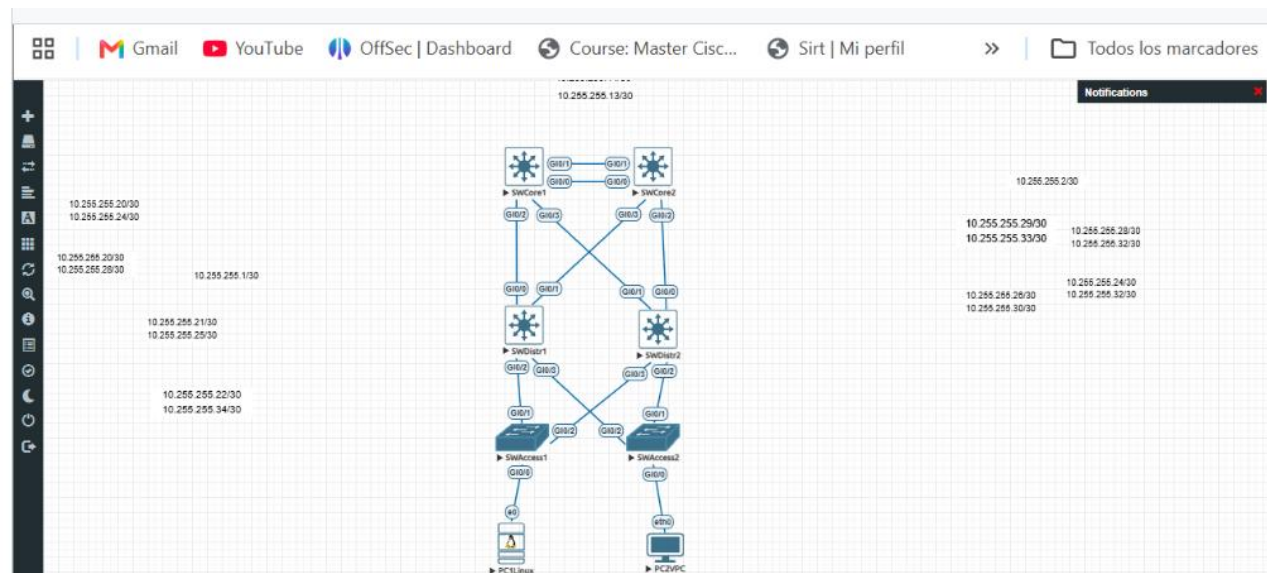


LAB03 LAN Jerárquica Tradicional Conmutada

Tecnologías: Inter-Vlan Routing, FHRP: HSRP, Dynamic Routing: OSPF, Rapid PVST, Etherchannel

Herramientas: Vmware, Eve-NG, Wireshark, Microsoft Visio, Visual Studio Code





Explorer: Lab03-STP,MSTP,Rapid PVST Configura..., Configuraciones, imagenes, swcore1-showipinterfacebrief.png, Topologia EVE-NG, Topologia Eve-NG Lab03.png, Topologia Logica, Lab03.vsd, Topologia Logica Lab03.png, Config-SWAccess1.ios, Config-SWAccess2.ios, Config-SWCore1.ios, Config-SWCore2.ios, Outline, Timeline

2.ios: README.md

```
4 # Herramientas: Vmware, Eve-NG,
23 ## Topología de la Red
29 ### Tabla de Direcccionamiento
58 | |G0/1|Enrutado|N/A|10.255.255.
26/30|10.255.255.25/30|SWCore1
(G0/3)|
59 | |G0/2|Trunk|99,1116,1132|N/A|
N/A|SWAccess2 (G0/1)|
60 | |G0/3|Trunk|99,1116,1132|N/A|
N/A|SWAccess1 (G0/2)|
61 | |Vlan 99|SVI|99|10.254.99.3/
24|HSRP 10.254.99.1/24 |N/A|
62 | |Vlan 1116|SVI|1116|10.254.16.
3/24|HSRP 10.254.16.1/24|N/A|
63 | |Vlan 1132|SVI|1132|10.254.32.
3/24|HSRP 10.254.32.1/24 |N/A|
64 |SWAccess1|G0/0|Access|1116|N/A|
```

Chat: Build with Agent. AI responses may be inaccurate. Generate Agent Instructions to. Tip: Use /create-agent to scaffold a custom agent for your workflow. + README.md. Describe what to b... + Local

SuperPuTTY - SWCore2

File View Tools Help

Protocol SSH Host Login Password Session

Commands

SWCore2 SWDist1 SWDist2 SWAccess1 SWAccess2

```
*Sep 26 19:38:39.273: %SYS-5-CONFIG_I: Configured from console by console
```

Opened session: CLI/192.168.0.200 [Telnet]

1.4.0.9 <LayoutName>

Descripción

Objetivo: Configurar la red, comprobar el comportamiento del tráfico en una topología jerárquica, configurar y hacer troubleshooting de los protocolos utilizados.

Protocolos y Tecnologías trabajadas: Inter-Vlan Routing, FHRP: HSRP, Dynamic Routing: OSPF, Rapid PVST

1. Creamos nuestros scripts de configuración en Visual Studio para cada uno de los dispositivos y configuramos desde la Capa de Acceso hacia las capas superiores.
2. Encendemos equipos desde la Capa de Core hacia las capas inferiores.
3. Comprobamos configuraciones con comandos show.
4. Comprobamos comportamiento de Rapid PVST, OSPF y HSRP.

Notas de campo

- En la topología tradicional de 3 capas se pueden usar varios diseños, en este caso se usa el diseño conmutado el cual la comunicación L3 se concentra entre capas de Core y Distribución a diferencia el diseño enrutado donde la comunicación L3 llega hasta la capa de acceso.
- Etherchannel: Empezamos por las interfaces físicas a agrupar en un portchannel y las apagamos primero, si el etherchannel es de L3 aplicamos no switchport , y luego aplicamos channel-group, de esta forma el Port Channel habrá recibido la configuración de las interfaces físicas y habremos evitado bucles en la creación de este.
- Las prioridades del protocolo HSRP o VRRP deben coincidir con el protocolo STP si estamos balanceando las VLANS, es decir ejemplo: si para SW1 esta activo el VIP del HSRP para vlan 10, entonces ese SW1 debe ser root primary STP para esa vlan.

Topología de la Red

Microsoft Visio

Topología Lógica Lab03

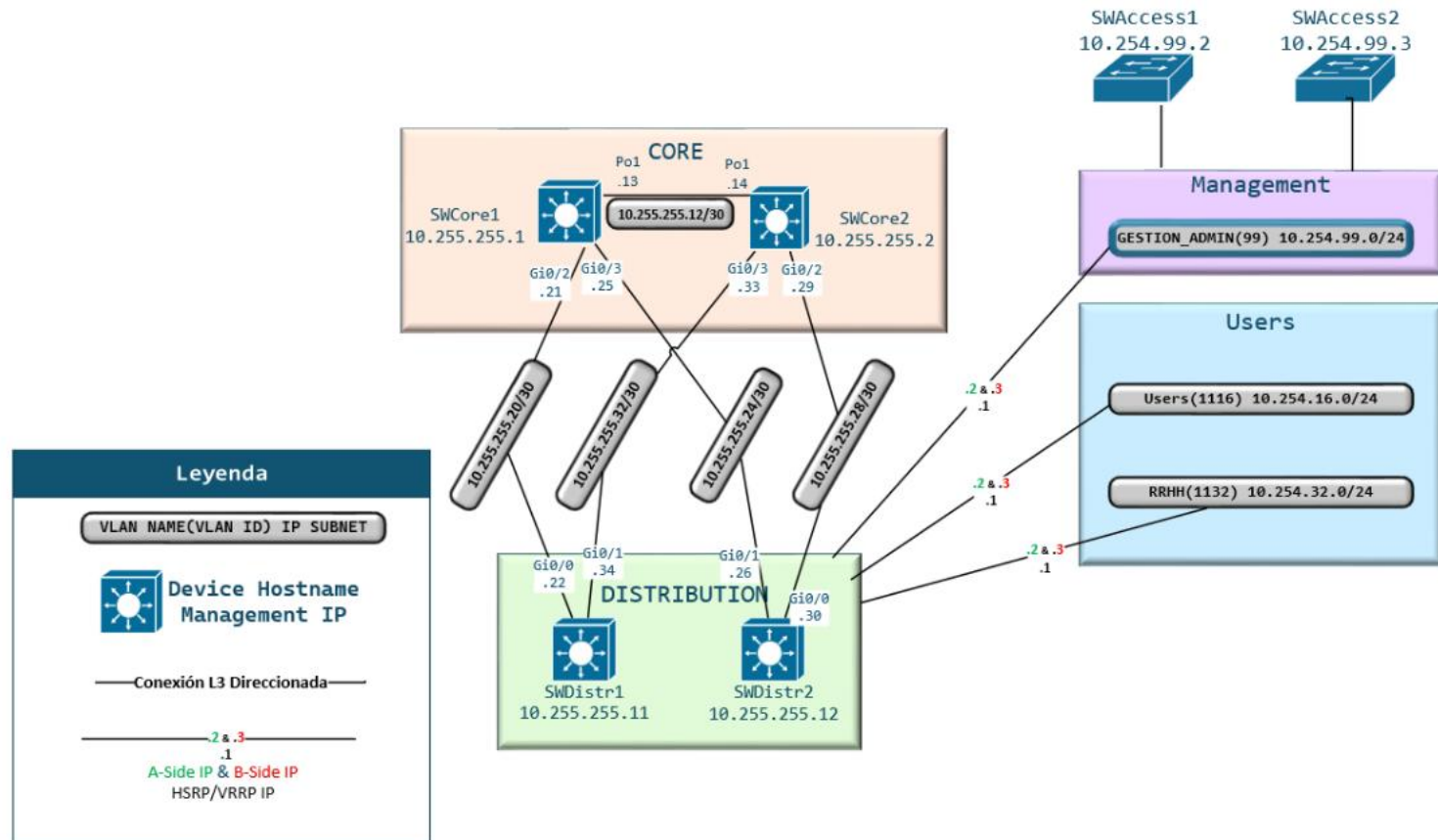


Tabla de Direccionamiento

Dispositivo	Interfaz	Tipo de Puerto	VLAN	Dirección IP / Máscara	Gateway / Next-Hop	Vecino/ Puerto Destino
SWCore1	Loopback 0	Logico	N/A	10.255.255.1/32	N/A	N/A
	Po1	Enrutado	N/A	10.255.255.13/30	10.255.255.14/30	SWCore2 (Po1)
	G0/0	Po1	N/A	N/A	N/A	SWCore2 (G0/0)
	G0/1	Po1	N/A	N/A	N/A	SWCore2 (G0/1)
	G0/2	Enrutado	N/A	10.255.255.21/30	10.255.255.22/30	SWDistr1 (G0/0)
	G0/3	Enrutado	N/A	10.255.255.25/30	10.255.255.26/30	SWDistr2 (G0/1)
SWCore2	Loopback 0	Logico	N/A	10.255.255.2/32	N/A	N/A
	Po1	Enrutado	N/A	10.255.255.14/30	10.255.255.13/30	SWCore1 (Po1)
	G0/0	Po1	N/A	N/A	N/A	SWCore1 (G0/0)
	G0/1	Po1	N/A	N/A	N/A	SWCore1 (G0/1)
	G0/2	Enrutado	N/A	10.255.255.29/30	10.255.255.30/30	SWDistr2 (G0/0)
	G0/3	Enrutado	N/A	10.255.255.33/30	10.255.255.34/30	SWDistr1 (G0/1)
SWAccess1	G0/0	Access	1116	N/A	N/A	PC1 Linux
	G0/1	Trunk	99,1116,1132	N/A	N/A	SWDistr1 (G0/2)
	G0/2	Trunk	99,1116,1132	N/A	N/A	SWDistr2 (G0/3)
SWAccess2	G0/0	Access	1132	N/A	N/A	PC2 VPC
	G0/1	Trunk	99,1116,1132	N/A	N/A	SWDistr2 (G0/2)
	G0/2	Trunk	99,1116,1132	N/A	N/A	SWDistr1(G0/3)

SWDistr1	Loopback 0	Logico	N/A	10.255.255.11/32	N/A	N/A
	G0/0	Enrutado	N/A	10.255.255.22/30	10.255.255.21/30	SWCore1 (G0/2)
	G0/1	Enrutado	N/A	10.255.255.34/30	10.255.255.33/30	SWCore2 (G0/3)
	G0/2	Trunk	99,1116,1132	N/A	N/A	SWAccess1 (G0/1)
	G0/3	Trunk	99,1116,1132	N/A	N/A	SWAccess2 (G0/2)
	Vlan 99	SVI	99	10.254.99.2/24	HSRP 10.254.99.1/24	N/A
	Vlan 1116	SVI	1116	10.254.16.2/24	HSRP 10.254.16.1/24	N/A
	Vlan 1132	SVI	1132	10.254.32.2/24	HSRP 10.254.32.1/24	N/A
SWDistr2	Loopback 0	Logico	N/A	10.255.255.12/32	N/A	N/A
	G0/0	Enrutado	N/A	10.255.255.30/30	10.255.255.29/30	SWCore2 (G0/2)
	G0/1	Enrutado	N/A	10.255.255.26/30	10.255.255.25/30	SWCore1 (G0/3)
	G0/2	Trunk	99,1116,1132	N/A	N/A	SWAccess2 (G0/1)
	G0/3	Trunk	99,1116,1132	N/A	N/A	SWAccess1 (G0/2)
	Vlan 99	SVI	99	10.254.99.3/24	HSRP 10.254.99.1/24	N/A
	Vlan 1116	SVI	1116	10.254.16.3/24	HSRP 10.254.16.1/24	N/A
	Vlan 1132	SVI	1132	10.254.32.3/24	HSRP 10.254.32.1/24	N/A

Pruebas y Verificación

Verificación de Configuración (show commands)

Tablas de enrutamiento

- Se han formado todas las adyacencias OSPF con éxito y los neighbors se han distribuido las rutas correctamente.

```
SWCore1 SWCore2 SWDistr1 SWDistr2 SWAccess1 SWAccess2
+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 11 subnets, 3 masks
O   10.254.16.0/24
    [110/2] via 10.255.255.26, 00:17:25, GigabitEthernet0/3
    [110/2] via 10.255.255.22, 00:17:35, GigabitEthernet0/2
O   10.254.32.0/24
    [110/2] via 10.255.255.26, 00:17:25, GigabitEthernet0/3
    [110/2] via 10.255.255.22, 00:17:35, GigabitEthernet0/2
O   10.255.255.2/32
    [110/3] via 10.255.255.26, 00:16:48, GigabitEthernet0/3
    [110/3] via 10.255.255.22, 00:16:48, GigabitEthernet0/2
O   10.255.255.12/30
    [110/3] via 10.255.255.26, 00:16:48, GigabitEthernet0/3
    [110/3] via 10.255.255.22, 00:16:48, GigabitEthernet0/2
O   10.255.255.28/30
    [110/2] via 10.255.255.26, 00:17:25, GigabitEthernet0/3
O   10.255.255.32/30
    [110/2] via 10.255.255.22, 00:17:35, GigabitEthernet0/2
SWCore1#
```

```
SWCore1 SWCore2 SWDistr1 SWDistr2 SWAccess1 SWAccess2
ia - IS-IS inter area, * - candidate default, U - per-user sta
o - ODR, P - periodic downloaded static route, H - NHRP, 1 - L
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 12 subnets, 3 masks
O   10.254.16.0/24
    [110/2] via 10.255.255.34, 00:13:51, GigabitEthernet0/3
    [110/2] via 10.255.255.30, 00:13:41, GigabitEthernet0/2
O   10.254.32.0/24
    [110/2] via 10.255.255.34, 00:13:51, GigabitEthernet0/3
    [110/2] via 10.255.255.30, 00:13:41, GigabitEthernet0/2
O   10.255.255.1/32
    [110/3] via 10.255.255.34, 00:13:41, GigabitEthernet0/3
    [110/3] via 10.255.255.30, 00:13:31, GigabitEthernet0/2
O   10.255.255.20/30
    [110/2] via 10.255.255.34, 00:13:41, GigabitEthernet0/3
O   10.255.255.24/30
    [110/2] via 10.255.255.30, 00:13:31, GigabitEthernet0/2
SWCore2#
```

```
SWCore1 SWCore2 SWDistr1 SWDistr2 SWAccess1 SWAccess2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
ia - IS-IS inter area, * - candidate default, U - per-user sta
o - ODR, P - periodic downloaded static route, H - NHRP, 1 - L
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 16 subnets, 3 masks
O   10.255.255.1/32
    [110/2] via 10.255.255.21, 00:14:50, GigabitEthernet0/0
O   10.255.255.2/32
    [110/2] via 10.255.255.33, 00:15:00, GigabitEthernet0/1
O   10.255.255.12/30
    [110/2] via 10.255.255.33, 00:15:00, GigabitEthernet0/1
    [110/2] via 10.255.255.21, 00:14:50, GigabitEthernet0/0
O   10.255.255.24/30
    [110/2] via 10.255.255.21, 00:14:50, GigabitEthernet0/0
O   10.255.255.28/30
    [110/2] via 10.255.255.33, 00:15:00, GigabitEthernet0/1
SWDistr1#
```

```
SWCore1 SWCore2 SWDistr1 SWDistr2 SWAccess1 SWAccess2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
ia - IS-IS inter area, * - candidate default, U - per-user sta
o - ODR, P - periodic downloaded static route, H - NHRP, 1 - L
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 16 subnets, 3 masks
O   10.255.255.1/32
    [110/2] via 10.255.255.25, 00:15:22, GigabitEthernet0/1
O   10.255.255.2/32
    [110/2] via 10.255.255.29, 00:15:22, GigabitEthernet0/0
O   10.255.255.12/30
    [110/2] via 10.255.255.29, 00:15:22, GigabitEthernet0/0
    [110/2] via 10.255.255.25, 00:15:22, GigabitEthernet0/1
O   10.255.255.20/30
    [110/2] via 10.255.255.25, 00:15:22, GigabitEthernet0/1
O   10.255.255.32/30
    [110/2] via 10.255.255.29, 00:15:22, GigabitEthernet0/0
SWDistr2#
```

Protocolo FHRP entre Switches de distribución

```
SWCore1 SWCore2 SWDistr1 SWDistr2 SWAccess1 SWAccess2
* Prohibited.
*****

User Access Verification

Username:
Username: admin
Password:
-----
[ ACCESO CONCEDIDO ]
Dispositivo actual: SWDistr1
Entorno: Eve-NG - Lab Win
-----

SWDistr1#show standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri P State  Active        Standby        Virtual IP
Vl99       1    110 P Active  local         10.254.99.3    10.254.99.1
Vl1116     1    110 P Active  local         10.254.16.3   10.254.16.1
Vl1132     1    100 P Standby 10.254.32.3   local         10.254.32.1
SWDistr1#
```

```
SWCore1 SWCore2 SWDistr1 SWDistr2 SWAccess1 SWAccess2
*****

User Access Verification

Username: admin
Password:
-----
[ ACCESO CONCEDIDO ]
Dispositivo actual: SWDistr2
Entorno: Eve-NG - Lab Win
-----

SWDistr2#show stand
SWDistr2#show standby bri
SWDistr2#show standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri P State  Active        Standby        Virtual IP
Vl99       1    100 P Standby 10.254.99.2   local         10.254.99.1
Vl1116     1    100 P Standby 10.254.16.2   local         10.254.16.1
Vl1132     1    110 P Active  local         10.254.32.2   10.254.32.1
SWDistr2#
```

Rapid PVST

*Se puede verificar que VLANs estan bloqueadas y cuales en Forwarding de cada switch.

```
SWCore1 SWCore2 SWDistr1 SWDistr2 SWAccess1 SWAccess2
SWDistr1#show spanning-tree summar
SWDistr1#show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: VLAN0099, VLAN1116
Extended system ID      is enabled
Portfast Default        is disabled
Portfast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default       is disabled
EtherChannel misconfig guard is enabled
Configured Pathcost method used is short
UplinkFast              is disabled
BackboneFast            is disabled

Name                    Blocking Listening Learning Forwarding STP Active
-----
VLAN0099                0          0          0          2          2
VLAN1116                0          0          0          2          2
VLAN1132                1          0          0          1          2
-----
3 vlans                 1          0          0          5          6
SWDistr1#
```

```
SWCore1 SWCore2 SWDistr1 SWDistr2 SWAccess1 SWAccess2
SWDistr2#show spanning-tree summ
SWDistr2#show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: VLAN1132
Extended system ID      is enabled
Portfast Default        is disabled
Portfast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default       is disabled
EtherChannel misconfig guard is enabled
Configured Pathcost method used is short
UplinkFast              is disabled
BackboneFast            is disabled

Name                    Blocking Listening Learning Forwarding STP Active
-----
VLAN0099                1          0          0          1          2
VLAN1116                1          0          0          1          2
VLAN1132                0          0          0          2          2
-----
3 vlans                 2          0          0          4          6
SWDistr2#
```

OSPF Wireshark

- Podemos observar el intervalo por defecto de 10 segundos del Hello de OSPF
- Observamos los intercambios de LSA al encender la topología.

No.	Time	Source	Destination	Protocol	Length	Info
1	2026-09-27 00:22:41.1041252	10.255.255.22	224.0.0.5	OSPF	94	Hello Packet
2	2026-09-27 00:22:44.0331212	10.255.255.21	224.0.0.5	OSPF	94	Hello Packet
3	2026-09-27 00:22:44.0657282	50:00:00:02:00:00	Nearest-Bridge	LLDP	371	MA/50:00:00:02:00:00 IN/Gi0/0 120 SysN=SMOistri1.winlab
4	2026-09-27 00:22:46.3275952	50:00:00:06:00:02	Nearest-Bridge	LLDP	373	MA/50:00:00:06:00:00 IN/Gi0/2 120 SysN=SMCore1.winlab
5	2026-09-27 00:22:50.3510502	10.255.255.22	224.0.0.5	OSPF	94	Hello Packet
6	2026-09-27 00:22:53.1435372	10.255.255.21	224.0.0.5	OSPF	94	Hello Packet
7	2026-09-27 00:22:53.3897452	10.255.255.22	10.255.255.21	OSPF	78	DB Description
8	2026-09-27 00:22:55.4475912	10.255.255.21	10.255.255.22	OSPF	78	DB Description
9	2026-09-27 00:22:58.0044522	10.255.255.22	10.255.255.21	OSPF	78	DB Description
10	2026-09-27 00:22:58.0052752	10.255.255.21	10.255.255.22	OSPF	98	DB Description
11	2026-09-27 00:22:58.0055032	10.255.255.22	10.255.255.21	OSPF	98	DB Description
12	2026-09-27 00:22:58.0056412	10.255.255.21	10.255.255.22	OSPF	78	LS Request

Frame 1: Packet, 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface - Ethernet II, Src: 50:00:00:02:00:00 (50:00:00:02:00:00), Dst: IPv4mcast_05 (01:00:5e:00:00:05), Internet Protocol Version 4, Src: 10.255.255.22, Dst: 224.0.0.5

Open Shortest Path First

- > OSPF Header
 - > OSPF Hello Packet
 - Network Mask: 255.255.255.252
 - Hello Interval [sec]: 10
 - > Options: 0x12, (L) LLS Data block, (E) External Routing
 - Router Priority: 1
 - Router Dead Interval [sec]: 40
 - Designated Router: 0.0.0.0
 - Backup Designated Router: 0.0.0.0
 - Active Neighbor: 10.255.255.1
 - > OSPF LLS Data Block

PING desde PC1 Linux a PC VPC

```
localhost:~# ip addr add 10.254.16.10/24 dev eth0
localhost:~# ip route add default via 10.254.16.1 dev eth0
localhost:~# ping 10.254.32.10
PING 10.254.32.10 (10.254.32.10): 56 data bytes
64 bytes from 10.254.32.10: seq=1 ttl=63 time=28.203 ms
64 bytes from 10.254.32.10: seq=2 ttl=63 time=15.568 ms
64 bytes from 10.254.32.10: seq=3 ttl=63 time=16.083 ms
64 bytes from 10.254.32.10: seq=4 ttl=63 time=11.328 ms
64 bytes from 10.254.32.10: seq=5 ttl=63 time=12.077 ms
64 bytes from 10.254.32.10: seq=6 ttl=63 time=11.697 ms
64 bytes from 10.254.32.10: seq=7 ttl=63 time=22.420 ms
64 bytes from 10.254.32.10: seq=8 ttl=63 time=18.184 ms
```



Skills Learned

- Configuración Intervlan
- Configuración de enrutamiento dinámico (OSPF)
- Configuración de protocolos de redundancia, disponibilidad y alta tolerancia (HSRP), (LACP), Etherchannel.
- Conocimiento de diseño tradicional de redes jerárquico
- Conocimiento y capacidad para Troubleshooting de OSPF, HSRP, LACP.